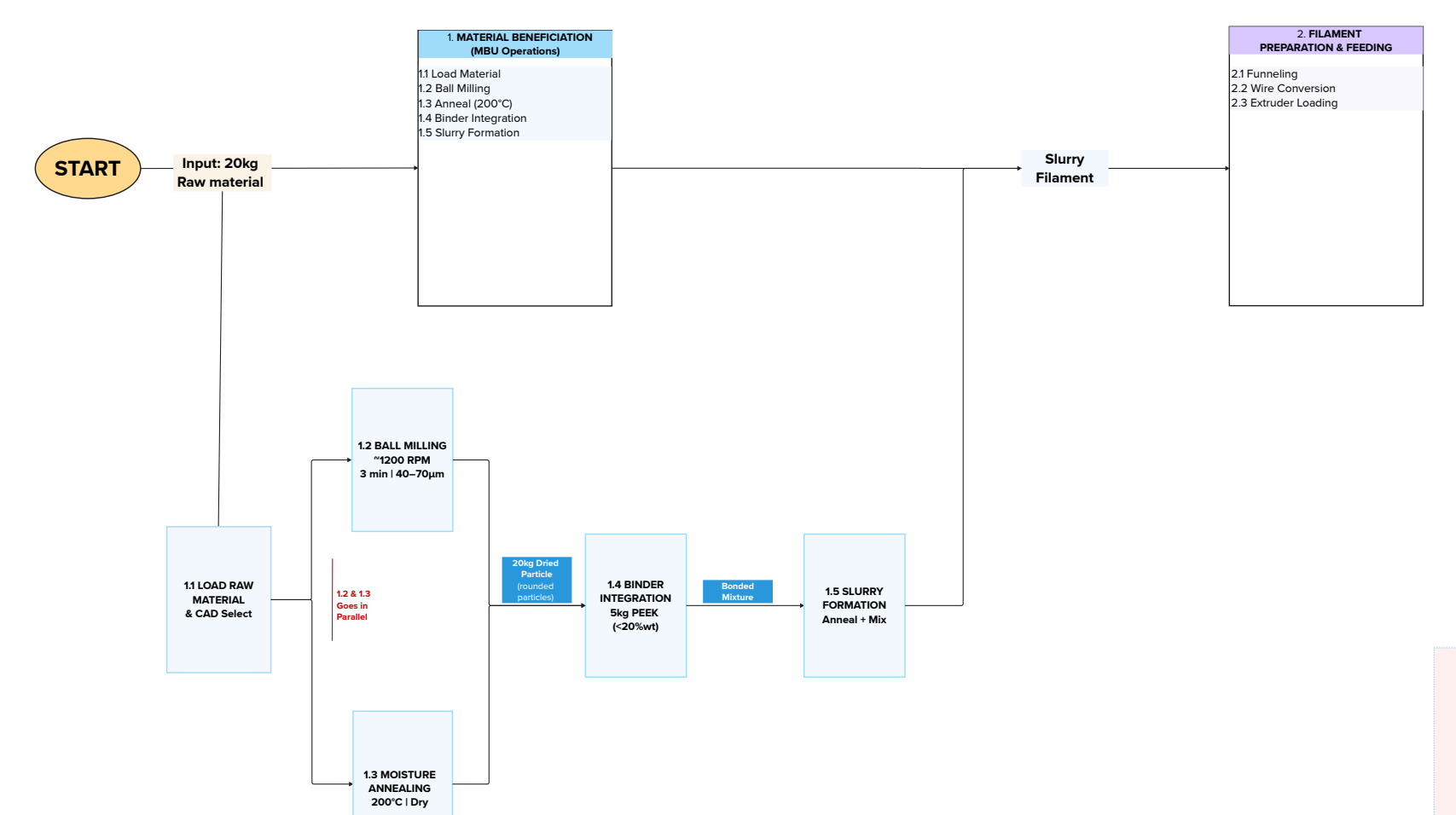
Tagging rules:
- [A], [B], [C]… — physical sensor readings / device state signals
- (software) — software inputs, file inputs, computed references (e.g. CAD file)
- (device) — device states that are not sensor readings (e.g. fan on/off, band heater zone state)
- *opt. — optional modules like LIBS/Raman

Chain continuity rule:
- Same letter carries through T1 → T2 → T3
- If a T2 item derives from two T1 inputs, it carries both letters: [A,B]
- If a T3 decision derives from a T2 item, it carries the same letter

See an example

AERIS: MATERIAL BENEFICIATION - DATA ARCHITECTURE


START
Input: 20kg Raw material
1. MATERIAL BENEFICIATION (MBU Operations)
1.1 Load Material
1.2 Ball Milling
1.3 Anneal (200°C)
1.4 Binder Integration
1.5 Slurry Formation
Slurry Filament
2. FILAMENT PREPARATION & FEEDING
2.1 Funneling
2.2 Wire Conversion
2.3 Extruder Loading
1.1 LOAD RAW MATERIAL & CAD Select
1.2 BALL MILLING ~1200 RPM 3 min | 40–70µm
1.2 & 1.3 Goes in Parallel
1.3 MOISTURE ANNEALING 200°C | Dry
20kg Dried Particle (rounded particles)
1.4 BINDER INTEGRATION 5kg PEEK (<20%wt)
Bonded Mixture
1.5 SLURRY FORMATION Anneal + Mix

SUBSYSTEM DEVICES & SENSORS
All components interfacing with this function block

1. Dual Centrifuge Mixer & Sieve (**Device**)
(MBU Module Capacity: ≥20kg)

2. Ball Mill Mechanism (**Device**)
(~1200 RPM | Output: 40–70µm)

3. Band Heater & Thermocouple Sensor (**Sensor**)
Target: 200°C & Moisture removal

4. Binder Intake Chamber (**Device**)
PEEK Pellets | ≤20% total weight

5. Funnel System (Transfer Channel) (**Device**)
Slurry Feed Line | To Filament Maker

6. Raman/LIBS Spectroscopy (**Sensor**)
(Optional Module) | Material ID & QC

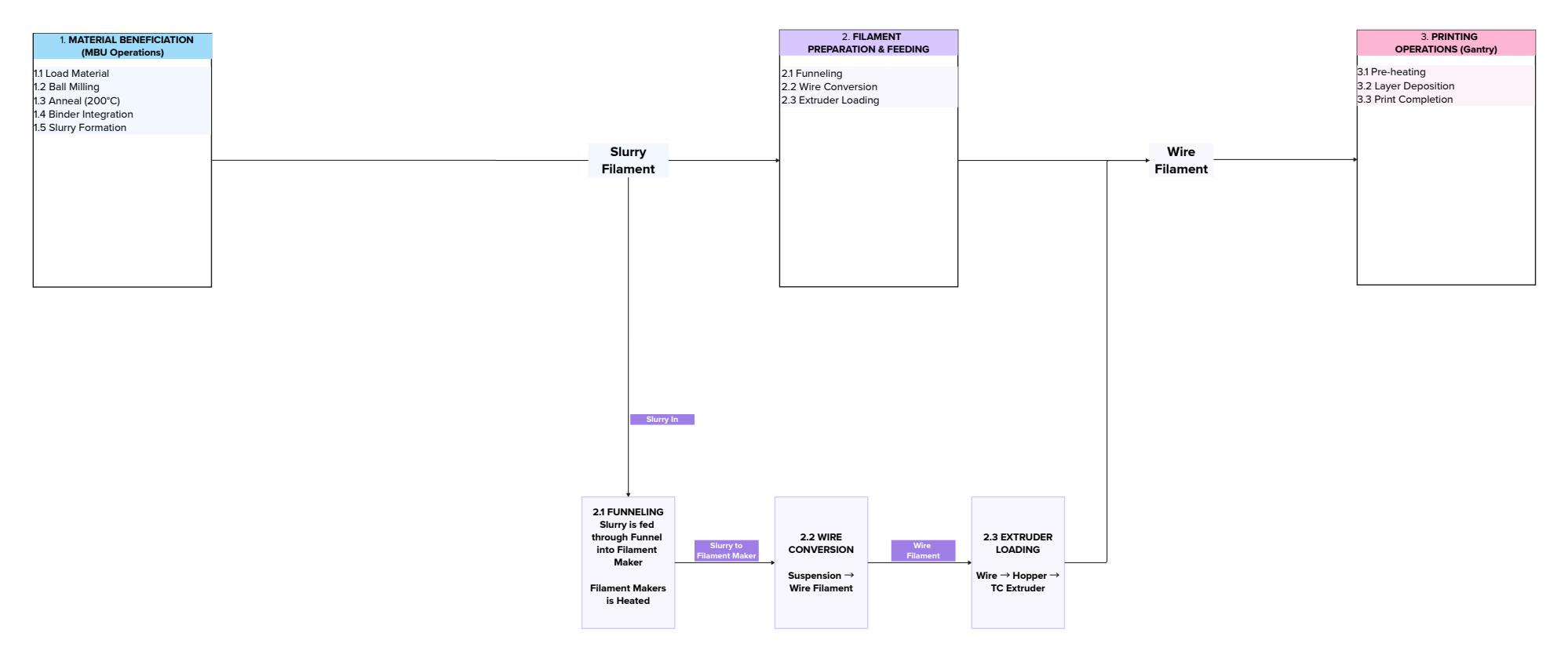
7. ML Camera System (MBU) (**Sensor**)
High-Speed | Defect Detection

IN	1.1 Load raw material & CAD select	1.2 Ball milling	1.3 Moisture annealing	1.4 Binder integration	1.5 Slurry formation	SLURRY FILAMENT
TIER 1 · RAW	[A] Raw mass signal [B] CAD file (software) [C] ML camera raw frames [D] Sieve state signal	[A] RPM signal (~1200 RPM) [B] TC voltage raw signal [C] Run time (min)	[A] TC raw temp °C [B] Output mass (20 kg) [C] Drying duration	[A] PEEK mass raw signal [B] Chamber state (device) [C] ML camera raw frames [D] Mix duration signal	[A] LIBS/Raman raw signal *opt. [B] ML camera raw frames [C] Funnel flow state (device) [D] TC raw temp °C	
TIER 2 · PROCESSED	[A] Confirmed mass vs ≥20 kg [B] CAD parameters parsed [C] Defect flag (ML cam) [D] Sieve pass confirmed	[A] Particle size dist. 40–70 µm [B] Temp stability check [A,C] RPM stability check [A,C] Output mass confirmed	[A] 200°C sustained [A] Moisture removed confirmed [B] 20 kg dried output confirmed	[A] PEEK wt% calculated [A] ≤20% verified [A,D] Mix uniformity [A,D] Bonded mixture confirmed	[A] Material ID (LIBS/Raman) *opt. [A] Elemental composition result *opt. [B] Defect map (ML cam) [B] Slurry consistency	
TIER 3 · DECISION	[A] Proceed to milling [A] Reject material [B] CAD mismatch alert [C] Re-inspect trigger — defect detected	[A] Continue cycle [A] Extend run time [A] Adjust RPM [A,C] Pass to anneal	[A] Trigger re-anneal [A] Trigger moisture alert [B] Hold — mass not confirmed [A,B,C] Pass to binder integration	[A] Approve mixture [A] Flag excess PEEK [A,D] Pass to slurry formation [A] Hold for recheck	[A] Log material ID (if LIBS active) [B] Route to rework — defect [B] Hold for QC [C,D] Release to filament fab	

See an example

AERIS: FILAMENT PREPARATION & FEEDING - DATA ARCHITECTURE

See an example


1. MATERIAL BENEFICIATION (MBU Operations)
1.1 Load Material
1.2 Ball Milling
1.3 Anneal (200°C)
1.4 Binder Integration
1.5 Slurry Formation
Slurry Filament
2. FILAMENT PREPARATION & FEEDING
2.1 Funneling
2.2 Wire Conversion
2.3 Extruder Loading
Wire Filament
3. PRINTING OPERATIONS (Gantry)
3.1 Pre-heating
3.2 Layer Deposition
3.3 Print Completion
Slurry In
2.1 FUNNELING Slurry is fed through Funnel into Filament Maker
Filament Makers is Heated
Slurry to Filament Maker
2.2 WIRE CONVERSION
Suspension → Wire Filament
Wire Filament
2.3 EXTRUDER LOADING
Wire → Hopper → TC Extruder

SUBSYSTEM DEVICES & SENSORS
All components interfacing with this function

1. Funnel System For Slurry Transfer (**Device**)
Gravity/Pressurized | Feed into Maker

2. Heated Filament Maker (**Device**)
Conversion Unit | Temp-Controlled

3. Band Heater & Thermocouple Sensor (**Sensor**)
Filament Temp | Monitor (TC)

4. Tungsten Carbide Extruder (**Device**)
Anti-Agglomeration | Nozzle + Barrel

5. Hopper (Pellet Feed) (**Device**)
PEEK Pellets | Inlet Port

6. Stepper Motor & Planetary Gearbox (**Device**)
Filament Drive | Speed Control

7. Aluminum Extrusion Mount (**Device**)
Structural Extruder Housing

8. Extruder Screw for Feed Mechanism (**Device**)
Positive Feed | Anti-Slip

9. Sheet Metal Cover & Fan (**Device**)
Heat Dissipation | Air Cooling

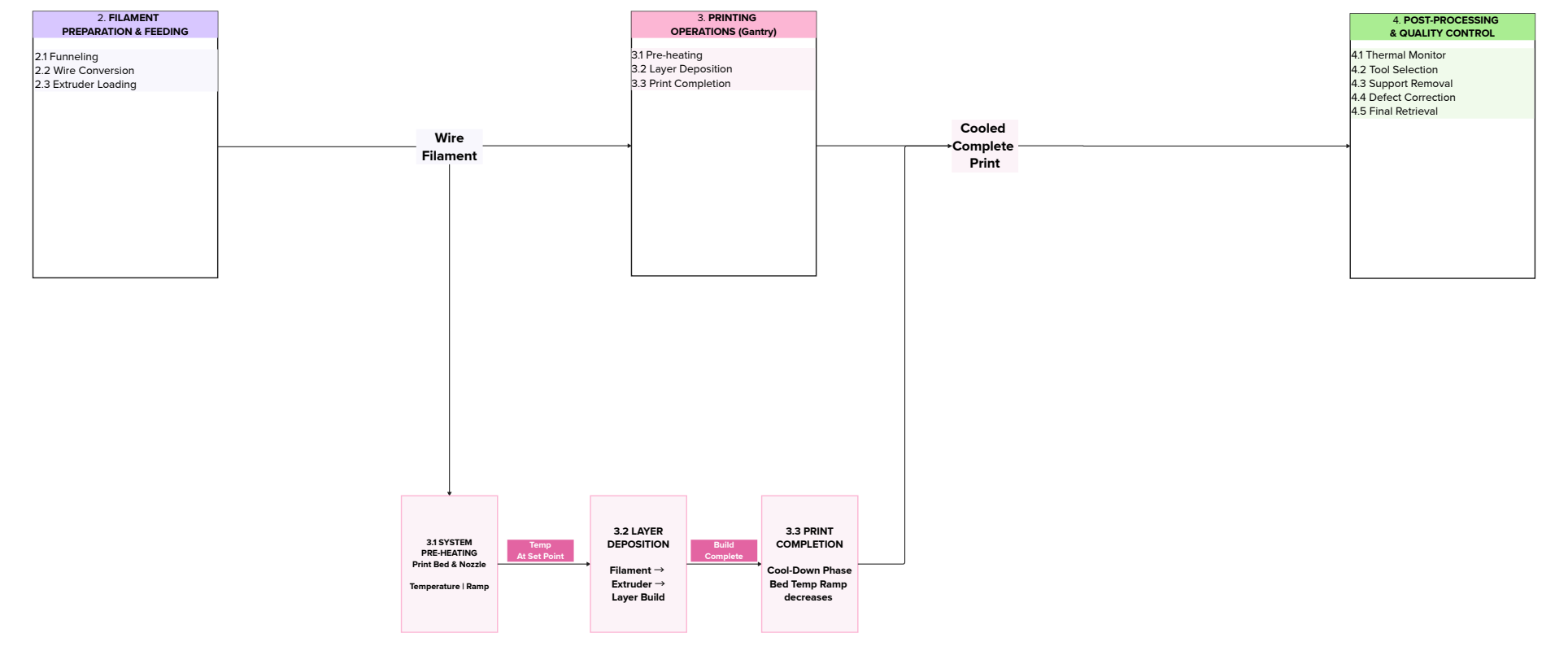
10. Coupling for Motor-Screw (**Device**)
Mechanical Transmission

11. Cables Out for Power & Data
Stepper Power | Encoder Feedback

SLURRY FILAMENT	2.1 Funneling	2.2 Wire Conversion	2.3 Extruder Loading	WIRE FILAMENT
TIER 1 · RAW	[A] Funnel flow state (device) [B] TC raw temp °C (band heater) [C] Slurry-in signal [D] Filament maker heated state (device) [E] Pressure mode: gravity / pressurized (device)	[A] TC raw temp °C (band heater) [B] Stepper motor encoder pulses [C] Planetary gearbox speed signal [D] Fan state (device)	[A] Hopper pellet presence signal [B] Extruder screw feed signal [C] Coupling torque signal [D] Encoder feedback — stepper (cables out)	
TIER 2 · PROCESSED	[A,C] Slurry feed rate [B,D] Filament maker temp stability [B,D] Heated state confirmed [A,E] Funnel clog check	[A] Temp sustained at operating level [B,C] Wire conversion rate [B,C] Drive speed computed [B,C] Filament diameter consistency *inferred [D] Cooling adequacy confirmed	[A] Hopper fill level confirmed [B,C] Feed continuity confirmed (anti-slip check) [B,C,D] Extrusion load [B,C,D] Wire output rate [B,C,D] Thermal uniformity — barrel & nozzle (TC extruder)	
TIER 3 · DECISION	[A,C] Continue feed [A,E] Switch pressure mode [A] Trigger clog alert [B] Trigger temp deviation alert [A,B,C] Pass to wire conversion	[A] Trigger temp deviation alert [B,C] Wire quality pass / fail [B,C] Adjust speed correction [D] Trigger cooling fault alert [A,B,C,D] Pass to extruder loading	[A] Trigger hopper refill [B,C] Trigger anti-slip alert [B,C,D] Pass wire filament to SS3 [B,C,D] Trigger jam flag [B,C,D] Extruder halt	

See an example

AERIS: PRINTING OPERATIONS (Gantry) - DATA ARCHITECTURE


2. FILAMENT PREPARATION & FEEDING
2.1 Funneling
2.2 Wire Conversion
2.3 Extruder Loading
Wire Filament
3. PRINTING OPERATIONS (Gantry)
3.1 Pre-heating
3.2 Layer Deposition
3.3 Print Completion
Cooled Complete Print
4. POST-PROCESSING & QUALITY CONTROL
4.1 Thermal Monitor
4.2 Tool Selection
4.3 Support Removal
4.4 Defect Correction
4.5 Final Retrieval
3.1 SYSTEM PRE-HEATING Print Bed & Nozzle
Temperature | Ramp
Temp At Set Point
3.2 LAYER DEPOSITION
Filament → Extruder → Layer Build
Build Complete
3.3 PRINT COMPLETION
Cool-Down Phase Bed Temp Ramp decreases

SUBSYSTEM DEVICES & SENSORS
All components interfacing with this function

1. 3D Printer Gantry System (**Device**)
XYZ Motion | Coordinate Control

2. Ceramic Build Plate (**Device**)
High-Temp Stable | Adhesion Surface

3. Anti-Agglomeration Extruder Nozzle (**Device**)
Tungsten Carbide Tip | Heated Barrel

4. Band Heaters for Nozzle & Barrel (**Device**)
Temp Setpoint | Zone Control

5. Thermocouples for Nozzle & Bed (**Sensor**)
Real-Time Temp Feedback | ±1°C

6. Vacuum Chamber for Print Enclosure (**Device**)
Oxidation Control | Inert Atmosphere

7. Crucible for melting material (**Device**)
High-Temp | Containment

8. ML Camera For Gantry (**Sensor**)
Layer Inspection | Real-Time QC

9. High-Speed Camera Array (**Sensor**)
Motion Capture | Defect Detection

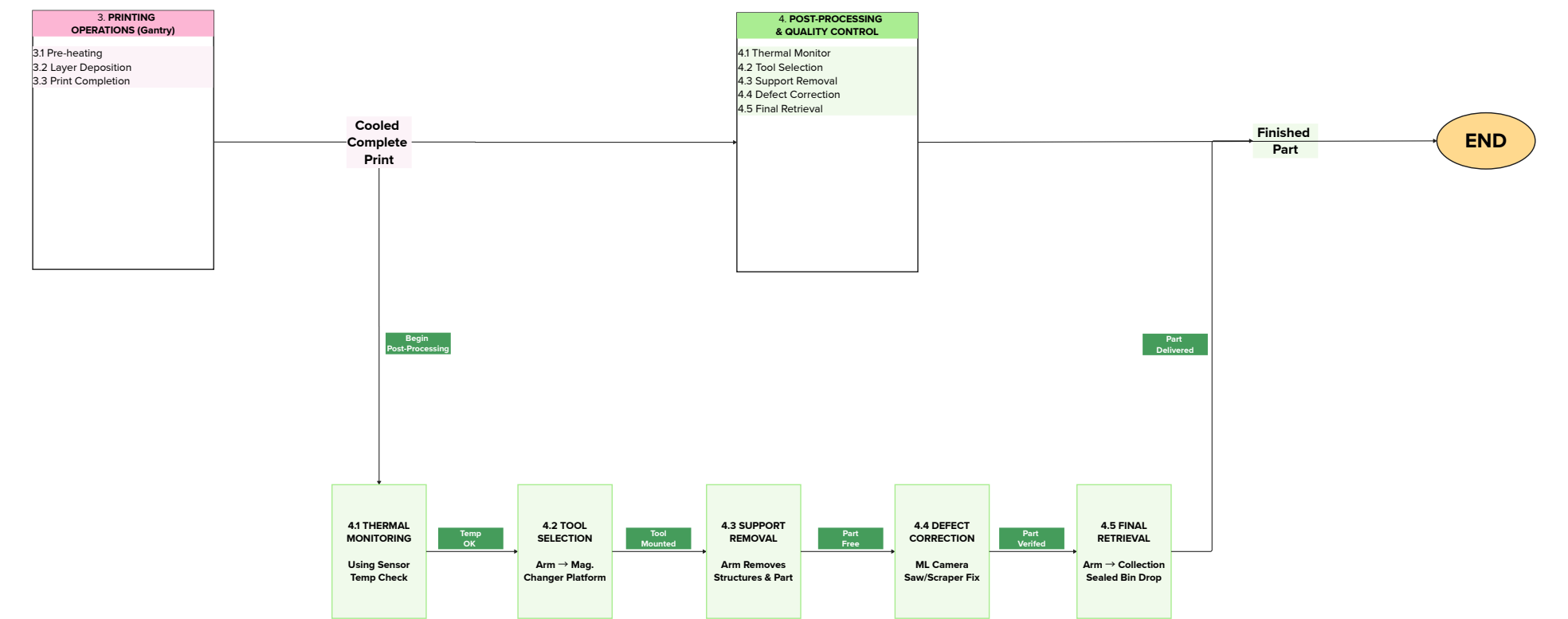
10. Ruggedized Interface Screen (**Device**)
Operator HMI | Status Display

11. Power System & Batteries
Peak: 3kW | Cont: 2.5kW

WIRE FILAMENT	3.1 System Pre-heating	3.2 Layer Deposition	3.3 Print Completion	COOLED COMPLETED PRINT
TIER 1 · RAW	[A] TC nozzle raw °C [B] TC bed raw °C [C] Band heater zone states (device) [D] Vacuum chamber pressure state (device)	[A] Gantry XYZ encoder position [B] TC nozzle & bed raw °C [C] ML camera raw frames (layer insp.) [D] High-speed camera raw frames [E] Crucible temp raw °C [F] Power draw raw (W)	[A] TC bed temp °C (ramp down) [B] TC nozzle temp °C [C] Power draw raw (W)	
TIER 2 · PROCESSED	[A] Temp ramp rate computed [A] Nozzle at setpoint confirmed [B] Bed at setpoint confirmed [D] Vacuum level confirmed [A,B] Power vs 2.5kW limit confirmed	[A] XYZ position vs CAD reference (software) [A] Layer height accuracy *from gantry encoder + ML cam [B] Thermal uniformity nozzle & bed [C,D] Defect map (ML + high-speed cam) [E] Crucible temp confirmed [F] Deposition rate	[A] Cool-down rate [A] Bed temp ramp curve computed [A,B] Part adhesion inferred (ceramic plate temp) [C] Power wind-down rate	
TIER 3 · DECISION	[A] Proceed to deposition [A] Halt — nozzle temp not reached [B] Halt — bed temp not reached [C] Trigger zone fault alert [D] Trigger vacuum insufficient alert	[A] Continue layer [A] Trigger position deviation alert [B] Trigger thermal deviation alert [C,D] Defect action: pause / correct [C,D] Layer pass / fail [A,B,C,D] Proceed to completion step	[A] Cool-down complete — release to SS4 [A,B] Trigger re-inspect [A,B] HMI operator notify [C] Power wind-down complete	

See an example

AERIS: POST-PROCESSING & QUALITY CONTROL - DATA ARCHITECTURE


3. PRINTING OPERATIONS (Gantry)
3.1 Pre-heating
3.2 Layer Deposition
3.3 Print Completion
Cooled Complete Print
4. POST-PROCESSING & QUALITY CONTROL
4.1 Thermal Monitor
4.2 Tool Selection
4.3 Support Removal
4.4 Defect Correction
4.5 Final Retrieval
Finished Part
END
Begin Post-Processing
Part Delivered
4.1 THERMAL MONITORING
Using Sensor Temp Check
Temp OK
4.2 TOOL SELECTION
Arm → Mag. Changer Platform
Tool Mounted
4.3 SUPPORT REMOVAL
Arm Removes Structures & Part
Part Free
4.4 DEFECT CORRECTION
ML Camera Saw/Scraper Fix
Part Verifed
4.5 FINAL RETRIEVAL
Arm → Collection Sealed Bin Drop

SUBSYSTEM DEVICES & SENSORS
All components interfacing with this function

1. Thermal Sensor Array (**Sensor**)
Print Chamber | Temp Monitoring

2. 7DOF Robotic Arm (**Device**)
Joint: Base/Shld/Elbow/Wrist | To control Yaw

3. Magnetic Tool Changer Platform (**Device**)
Auto End-Effector| Swap System

4. End Effector Suite: Gripper (**Device**)
Saw | Scraper Support Tools

5. ML-Reinforced Camera System (**Sensor**)
Delamination & Adhesion Detection

6. Sealed Collection Bin (**Device**)
Final Part Deposit Area

7. High-Speed Camera for QC (**Sensor**)
ML Defect ID | Correction Trigger

8. Trigger System (Cooling → Arm) (**Sensor**)
Temp Threshold | Activation Logic

9. Ceramic Build Plate (**Device**)
Part Adhesion | Release Surface

10. Scraper Tool End Effector (**Device**)
Surface Finish | Burr Removal

11. Saw Tool End Effector (**Device**)
Support Cut | Overhang Trim

12. Ruggedized Interface Screen (**Device**)
Status Display | User Alert

COOLED COMPLETED PRINT	4.1 Thermal Monitoring	4.2 Tool Selection	4.3 Support Removal	4.4 Defect Correction	4.5 Final Retrieval	FINISHED PART
TIER 1 · RAW	[A] Chamber temp raw °C [B] Trigger system threshold signal	[A] 7DOF joint angles raw [B] Magnetic tool changer state (device) [C] Gripper state (device)	[A] Arm joint angles raw [B] Saw/scraper contact signal (removal) [C] Ceramic plate adhesion state *inferred [D] Part position via arm encoder	[A] ML cam raw frames (delamination & adhesion) [B] High-speed cam raw frames [C] Saw/scraper contact signal (correction)	[A] Arm joint angles raw [B] Gripper state (device) [C] Sealed bin deposit signal	
TIER 2 · PROCESSED	[A,B] Chamber temp vs threshold [A,B] Temp OK confirmed [A,B] Activation logic resolved (trigger system)	[A] Arm trajectory computed [A,B] Tool selected & mounted confirmed [A] Joint position accuracy [A,C] End effector engagement confirmed	[A,B] Supports removed confirmed [A,B] Part free confirmed [A,B] Contact force computed [A,C] Adhesion release inferred (plate + arm force)	[A,B] Defect map (ML + high-speed cam) [A,B] Delamination detected Y/N [A,B] Surface correction measured	[A] Arm path to bin computed [B,C] Deposit confirmed [A,B,C] Part delivered status [C] Bin deposit position verified	
TIER 3 · DECISION	[A,B] Arm activate [A,B] Halt — temp not safe [A,B] Trigger operator alert via HMI	[A,B] Proceed with selected tool [A,B] Re-select tool [A,B] Trigger mount failure alert [A,B] HMI notify	[A,B] Part free — proceed [A,B] Re-attempt removal [A,C] Trigger plate release [A,D] Route to re-inspect — position error	[A,B] Pass to retrieval [A,B] Route to rework — defect confirmed [A,B,C] Apply correction (saw / scraper) [A,B] Correction complete — confirmed	[A,B,C] Bin drop confirmed — finished part [A,B,C] Bin miss — re-retrieve trigger [A,B,C] End cycle log to HMI [A,B,C] HMI alert issued	

See an example